

Subject	:	HCI & Computer Graphics	Section:	A, B, C	Date:	14th April 2026
Course Code	:	CSC-335			Marks:	33
Examination	:	Mid Term	Spring-2025		Semester :	6 th
Instructor	:	Hamna Anwar & Sania Khalid			Time:	1 hour

Note:

- 1- External Source and Use of mobile-phones during the exam-leads to Disciplinary Action.
- 2- Plagiarism or copying from others will result in disqualification from the exam.
- 3- There are total of 8 questions and 7 pages in this question paper.
- 4- where required, write 1 to 2 sentence explanations, instead of long paragraphs. Failure to comply will result in marks deduction.
- 5- Answer all questions on the answer booklet.

CLOs	Questions
CLO1: (C2: Understand) Explain fundamental principles of user-centered design	2, 3
CLO2: (C2: Understand) Identify accessibility & inclusivity as essential elements of HCI	7, 8
CLO3: (C4: Analyze) Analyze user feedback to refine interface prototypes	5, 6
CLO4: (C5: Design) Design interactive prototypes using modern HCI tools	1, 4

Q1: A user is trying to send Rs. 5,000 and accidentally presses Cancel instead of Confirm.

- a) Classify this error as a slip, mistake, or lapse. [1 mark]
- b) Identify **two** design problems with how the two buttons are displayed. [1 mark]
- c) Redesign the buttons to overcome the **two** problems identified in part b. [2 marks]

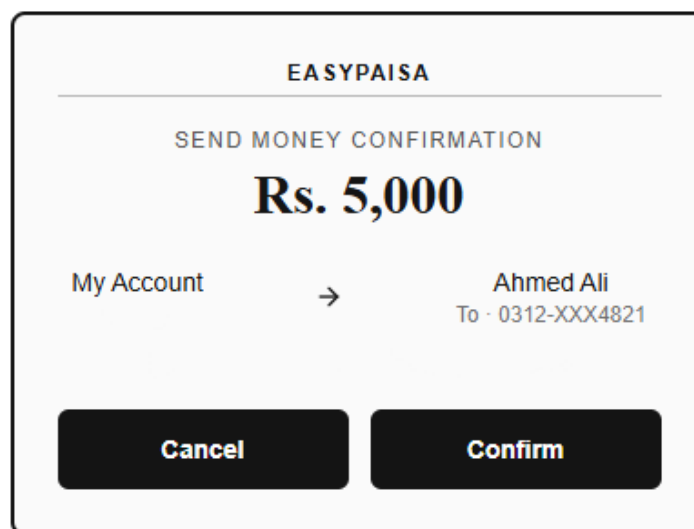


Fig 1. UI for Question 1

Q2: Ayesha submits her HCI Research Paper. She later notices an error and clicks "Delete" expecting the file to be permanently removed. Instead, it disappears from "Submitted" and reappears under the "Drafts" tab.

- Identify Ayesha's mental model and the system model in this scenario. Write one sentence for each. [2 marks]
- Would you fix this with a label change or a design change? [1 mark]

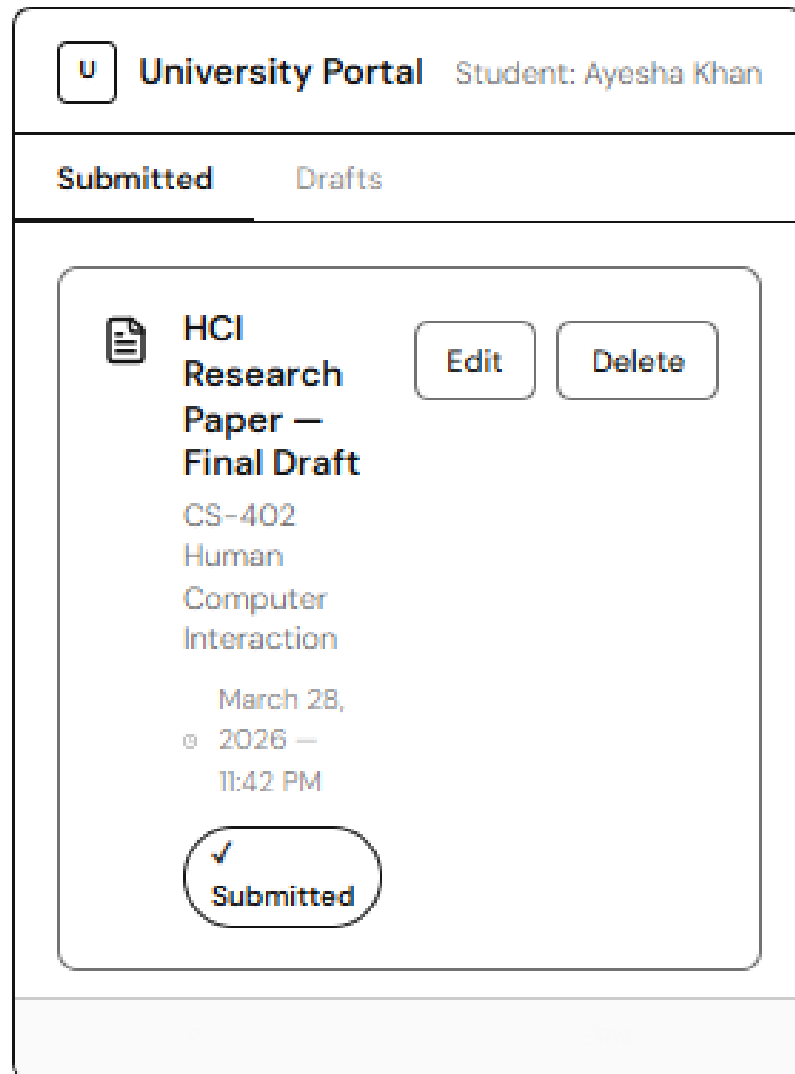


Fig 2 UI for Question 2

Q3: Look at the BNU Contact Directory screen below.

- A student wants to call Sara Khan but she reads the wrong number. Name the Gestalt principle which is causing this mistake. [1 mark]
- The student wants to call Usman Tariq using the Quick Actions section. She cannot tell which items she can tap and which are just text. Name the Gestalt principle being violated and identify what is missing from the "Call" buttons that would have prevented this confusion. [1 mark]

BNU Contact Directory	
Search contacts:	
Type a name...	
Faculty — Department of Computer Science	
Ahmed Ali	
0312-4471829	
Sara Khan	
0321-9983012	
Usman Tariq	
0333-5512678	
Fatima Malik	
0345-7723490	
Bilal Hassan	
0300-1192847	
Quick Actions	
Ahmed Ali	Call
Sara Khan	Call
Usman Tariq	Call
Fatima Malik	Call
Bilal Hassan	Call
BNU IT Department — For support contact helpdesk@bnu.edu.pk	

Figure 3. UI for Question 3

Q4: A user is placing an order on Daraz. Figure 4 shows the UI before pressing the “Place Order” button. Figure 5 shows the UI after pressing the “Place Order” button. Figure 6 shows the UI 5 seconds after pressing the “Place Order” button.

- Name the stage in the Norman's Action Cycle where a breakdown occurs and explain why in 1 to 2 sentences. [2 marks]
- Identify if this a Gulf of Execution or Gulf of Evaluation problem. **NO** need to explain the answer [1 mark]
- Suggest one design change that would resolve this issue. Show the redesigned UI. Do not explain the redesigned UI. [1 mark]

<

Daraz

Order Summary

Delivery Address

Ahmed Khan
House 12, Street 5, Gulberg III, Lahore
+92 312 456 7890

Order Items

Product Image

Wireless Bluetooth Earbuds — Noise Cancelling

Qty: 1

Rs. 2,499

Product Image

Phone Case — Transparent Slim Fit

Qty: 1

Rs. 350

Subtotal

Rs. 2,849

Delivery Fee

Rs. 99

Total

Rs. 2,948

Payment Method

Cash on Delivery

Place Order

Figure 4: UI before pressing "Place Order"

<

Daraz

Order Summary

Delivery Address

Ahmed Khan
House 12, Street 5, Gulberg III, Lahore
+92 312 456 7890

Order Items

Product Image

Wireless Bluetooth Earbuds — Noise Cancelling

Qty: 1

Rs. 2,499

Product Image

Phone Case — Transparent Slim Fit

Qty: 1

Rs. 350

Subtotal

Rs. 2,849

Delivery Fee

Rs. 99

Total

Rs. 2,948

Payment Method

Cash on Delivery

Place Order

Figure 5: UI after pressing "Place Order"

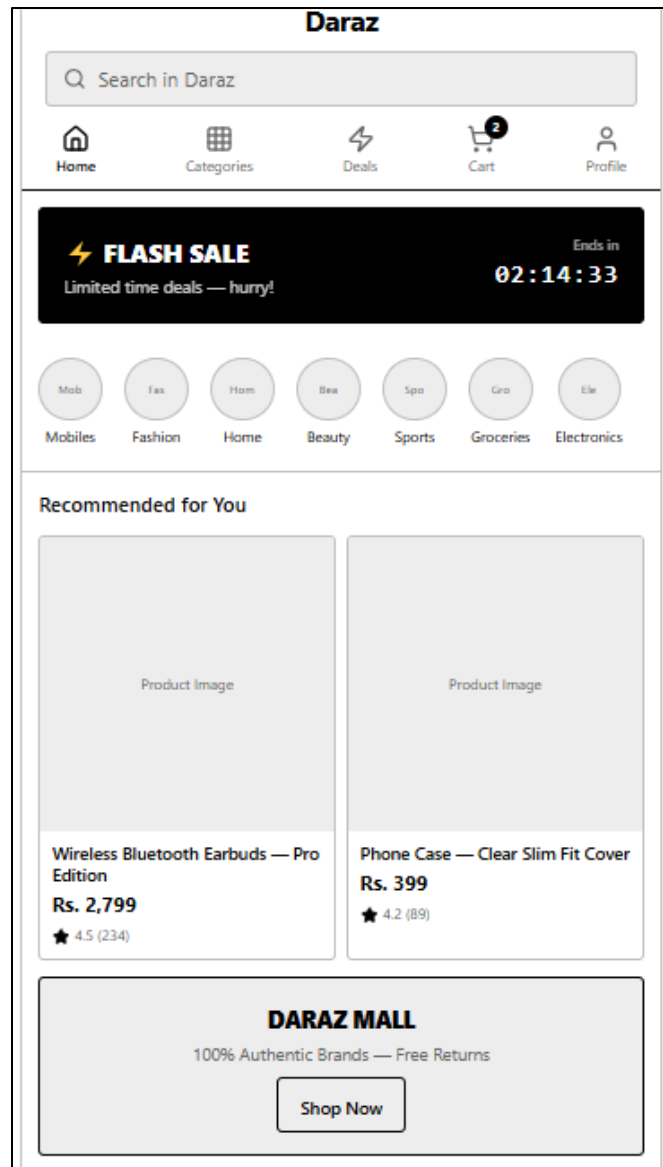


Figure 6. UI for Question 4: 5 seconds after pressing the “Place Order” button

Q5: [Answer each part of this question in 1 to 2 sentences only]

Rozee.pk redesigns their mobile app. The new version is visually polished and loads faster than before. During post-launch research, the UX team interviews several users and gets the following responses:

- i. "I didn't know Rozee had an app, I always just used the website."
 - ii. "I downloaded it but you have to link your LinkedIn before you can even browse jobs. I don't have LinkedIn so I just deleted it."
 - iii. "I use it every day but honestly the jobs it shows me are never in my field. I'm a graphic designer and it keeps suggesting call center jobs."
 - iv. "It's fine to use, I figured it out quickly enough."
- a) Three of these four responses reveal a UX problem. Identify which three using the Honeycomb facet. [3 marks]
 - b) The fourth response suggests one Honeycomb facet is working well. Identify the facet, and explain what this tells the team about where to focus their redesign effort. [2 marks]
 - c) The team decides to refine the app based on response ii. Suggest one specific change to the app that directly addresses this user's feedback. [1 mark]

Q6: [Answer each part of this question in 1 to 2 sentences only]

A UX team at a Pakistani e-commerce company has survey data showing that 73% of users abandon their cart at the payment step. The CEO says: "We have our answer, fix the payment step". Only one word or one-line answers are required.

- a) What does the survey data tell the team? [1 mark]
- b) What critical information does the survey fail to provide? [1 mark]

The team conducts user interviews. The most commonly reported issue across participants is: "I didn't realize my session had timed out, I thought my card details were saved but I had to start over."

- c) What research method should the team use next to confirm this is a widespread problem? [1 mark]
- d) Based on this feedback, suggest one specific change to the checkout screen that directly addresses what users experienced. [1 mark]

Q7: [Answer each part of this question in 1 to 2 sentences only]

A UX team is designing the ride request acceptance feature for an app similar to the inDrive app. To test it, they recruit 12 drivers through a university noticeboard, bring them into their office, seat them at in their meeting room with a desk, and ask them to use the app on a phone mounted in a stand. The session runs at 2pm on a Tuesday. All participants complete every task. The product manager reviews the report and approves the design for release.

- a) Identify two context factors the team failed to account for. [1 mark]
- b) For each factor you identified, state one design decision that would have been made differently had it been considered during the research phase. [1 mark]

- c) The team celebrated their user testing by saying: "We recruited actual drivers." A colleague reviews the recruitment method and disagrees. What exactly is the colleague's concern? [1 mark]
- d) Some drivers who use ride-hailing apps use the phone one-handed due to a physical limitation. Identify one design change that would make the ride acceptance screen more inclusive for these users. [1 mark]

Q8: [Answer each part of this question in 1 to 2 sentences only]

A university hires a design team to build a food ordering app for hospital patients. The team defines their user as anyone who wants to order food and spends three weeks making the app look modern and visually polished. They test it with 15 university students. All participants complete every task successfully and rate the app 4.8 out of 5.

The project manager presents the results to the hospital board and says: "The app looks great and scored 4.8, it's ready to ship."

- a) The project manager uses the 4.8 rating as evidence that the design works. Give two reasons why using the score as proof that the app is ready for hospital patients is wrong. [2 marks]
- b) If you are tasked to lead another team to work on the same problem, would you use the same approach as Team A? If not, mention two things you would have done. [2 marks]
- c) The stakeholder is impressed with the polished product produced by Team A and is inclined to select their design. What is the issue in the stakeholder's thinking? [1 mark]
- d) A hospital patient using this app may be elderly, have limited hand mobility, or be recovering from surgery. Identify one specific design decision the team should have made to ensure the app works for these users. [1 mark]